

HELIOLITE · PROBLEM STATEMENT

HelioLite problem statement

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-09-26	Amish Chadha	Stronger sources
0.3	2026-09-25	Amish Chadha	TRL 3 update; budget \$430 and no-sun-sensor constraint as decided by Amish (HLT-DDR-001); siting findings from HLT-CAL-001; open items listed; EN 17037 sunlight citation corrected
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (users, context, siting, constraints, out of scope, cited prior work)
0.1	2026-09-24	Amish Chadha	Initial scaffold

HelioLite problem statement

Rooms that face away from the equator (north-facing in the Northern Hemisphere) and the shaded back of many greenhouses never receive direct sun in winter, so they stay dim and cold while sunlight falls a few meters away. A small two-axis mirror that follows the sun can send that sunlight through a chosen window all day, but the heliostats on the market are either large power-plant or civic installations or discontinued consumer products. HelioLite aims to be a garage-buildable, open design at about \$455 in parts.

■ The problem

In the Northern Hemisphere a north-facing window sees only diffuse skylight. At 45° N the noon sun stands only about 21.6° above the horizon at the winter solstice (90° minus latitude minus 23.44°), so neighboring buildings, trees and the house itself also shade many south, east and west openings for much of the winter. The result is:

1. **Too little daylight.** People in the United States spend, on average, about 90 % of their time indoors (US EPA), so a room without winter sun leaves its occupants with little daylight for much of the season.
2. **No passive solar gain.** Those rooms get no free winter heat, and greenhouse beds on the north side grow slowly.
3. **Electric light as the fallback.** Occupants turn on lamps during the day.

A heliostat is a mirror turned by two motors so that the reflection of the sun stays on a fixed target as the sun moves. The physics is well proven. What is missing is a small, low-cost, repairable unit that a homeowner, school or community garden can build, site and calibrate without special tools, and that is safe to run next to people.

■ Users and context

User	Need	Context
Homeowner or tenant with a dark north room	Sunlight in a living room, kitchen or home office on clear winter days	Suburban yard or courtyard, target window 3 to 20 m away
Small greenhouse or community garden	Light and some heat on shaded beds in winter and spring	Hoop house or lean-to, target is a glazed end wall
School or makerspace	A teaching project in astronomy, optics and control that does real work	Rooftop or yard, supervised use
Maker who builds and maintains it	Buy common parts, print the gimbal, flash firmware, calibrate from a phone	Home workshop, hand tools, a 3D printer

Operating environment

- **Latitude:** 30° to 55° N (and the mirror image in the Southern Hemisphere). Winter sun elevations at noon from about 12° to 37°.
- **Sun:** direct normal irradiance (DNI) of about 700 to 900 W/m² on clear winter days; nothing useful on overcast days.
- **Weather:** outdoor all year, -20 to +45 °C ambient, rain, snow, hail, UV and wind gusts that can exceed 30 m/s (108 km/h) in storms.

- **Siting:** the mirror must see the sun during the useful hours and see the target. A unit placed directly north of a house must stand outside the house's winter shadow (about 15 m for 6 m eaves at 45° N), so many sites will use a side yard, a roof or a garage.
- **Power:** a 12 V supply fed from an indoor outlet; Wi-Fi often available but not guaranteed.
- **People:** the beam crosses gardens, paths and windows where people, pets, drivers and neighbors may look into it.

■ Constraints

- Garage-buildable prototype, about \$455 USD in parts (`project.yaml` budget, raised from \$400 to \$430 by Amish on 2026-09-25, HLT-DDR-001 D4, then to \$455 by a top-up he approved on 2026-09-26, HLT-DDR-002 P1).
- Open-loop tracking from time and location (a sun-position algorithm), with no sun sensor or camera. Homing switches for the axes and a weather anemometer for storm stow are allowed (HLT-DDR-001 D2, D5).
- Only extra-low voltage (12 V DC) outdoors; the mains adapter stays indoors.
- Parts available from general hardware, electronics and 3D-printing suppliers; one mirror cut to size by a local glazier.
- Install by two people with hand tools, without a crane or permanent roof penetration.
- Must not create a hazard to people, vehicles or aircraft from the reflected beam, including when stowed or unpowered.

■ Out of scope

- Concentrating sunlight for heating water, cooking or power (a flat mirror only redirects about one sun).
- Arrays of heliostats aimed at one target.
- Light pipes, fibers or daylight-redirecting films inside the building.
- Commercial product certification.

■ Prior work

- **Civic heliostats.** Since 2013, three computer-driven heliostats with 51 m² of mirror in total, about 450 m above the town of Rjukan, Norway, reflect sunlight onto an ellipse of about 600 m² in the market square during the months when the valley is in shadow ([Visit Rjukan](#)). The project shows the social value of redirected winter sun at a scale far above a house.
- **Home heliostats.** The Sunflower home heliostat was marketed around 2012 as a self-powered unit redirecting up to about half a square meter of sunlight, described as up to 500 W, into a home ([Inhabitat](#); [Solar Power World](#)). It confirms the use case; it does not appear to be widely available today, and its design is not open.
- **Open maker heliostats.** Open Arduino projects compute the sun's position from a clock and drive a mirror in altitude and azimuth, for example [jremington/Arduino_heliostat](#) (steppers with gear reduction, about 0.1° step resolution, and a warning that the Arduino's own crystal is not accurate enough for long unattended tracking) and the Cerebral Meltdown [Open Sun Harvesting Project](#). The [Open Source Ecology heliostat page](#) collects further examples. Heliolite builds on this approach but targets an outdoor-rated, calibrated, safe unit rather than a bench model.

- **Sun-position algorithms.** NREL's Solar Position Algorithm computes the sun's position with an uncertainty of $\pm 0.0003^\circ$ for the years -2000 to 6000 ([Reda and Andreas, NREL/TP-560-34302](#); [NREL MIDC SPA page](#)). Grena's five lighter algorithms, valid for 2010 to 2110, have maximum errors from 0.19° down to 0.0027° and suit small microcontrollers ([Grena 2012, Solar Energy 86](#)). Either is far more accurate than the mechanics of a low-cost unit.
- **Heliostat wind loads.** Wind-tunnel data and design methods for ground-mounted heliostats are given in Sandia report SAND92-7009 ([Peterka and Derickson 1992](#)) and updated by the Heliostat Consortium ([NREL 2025](#)). Heliolite uses these only for first-order estimates.
- **Glare and eye hazard.** Sandia's analytical method for glint and glare from solar mirrors relates retinal irradiance and subtended angle to after-image and retinal burn risk ([Ho, Ghanbari and Diver 2011](#)). It is the basis for the beam-safety requirements in HLT-REQ-001.

■ Open questions

- Which first site and user (a house with a north room, a greenhouse or a school)? This sets latitude, distance to target and siting. Proposed, awaiting Amish (HLT-DDR-001, O1); co-design partners stay open by Amish's instruction.
- How common is a site where the mirror can see the winter sun and the north window at the same time, without standing in the house's shadow? The hourly model in HLT-CAL-001 (section B) shows the answer matters more than any hardware choice: the same unit delivers 0.40 to 1.00 kWh on the solstice at three illustrative sites. A short siting survey of real yards is needed (O3).
- Is daylight or heat the main benefit users value? The calculations suggest daylight is substantial (about 500 lx) and heat is modest (about 150 W while sunny).
- What local rules on glare, setbacks and structures on a lot or roof apply at the first site (O2)?

■ User research and co-design

This design is for households and growers the author does not know yet, so requirements should be checked with them.

- [] Identify two or three candidate sites (house, greenhouse, school) and their owners
- [] Record latitude, target size and distance, shading, and where people walk
- [] Ask users whether they want light, heat or both, and for how many hours
- [] Revise requirements (REQ) from findings before freezing the design